

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		65608	
Total Threats Mitigated	23	Total Packets Analyzed		9206027971	
High Priority	12	Malicious Packets		9181702364	
Low Priority	11	True Negatives Passed		53616	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		0	
ML Metrics	Value	SYN Flood		0	
False Positives	32	UDP Flood		17	
FP Rate	0.05%	Uncertain		6	
IF Precision	99.77%				
IF Recall	93.11%	Action		Count	
IF F1-Score	96.32%	Quarantined		4	
IF Accuracy	98.61%	Time Ban		3	
		Blackhole		3	
		Released		6	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.77%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	93.11%	Share of actual attacks successfully flagged
F1-Score	96.32%	Balanced measure combining Precision and Recall
Accuracy	98.61%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.05%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	6.89%	Actual attacks that went undetected
True Positive Rate (TPR)	93.11%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.95%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 11935	FN 883
Actual: Normal	FP 28	TN 52733

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	11935

2c. Response Latency

Metric	Value	Description
Detection Time	351.6 ms	Average time taken to detect an attack
Mitigation Response Time	4.9 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	0.93%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	2.92%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	2.64%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.6	6	6	2026-09-11 11:12:43	2026-09-11 11:33:00
10.0.0.21	10	3	2026-09-11 11:15:56	2026-09-11 11:37:41
10.0.0.22	10	3	2026-09-11 11:15:57	2026-09-11 11:37:41
10.0.0.20	10	2	2026-09-11 11:15:56	2026-09-11 11:37:48
10.0.0.2	1	1	2026-09-11 11:22:38	2026-09-11 11:22:48
10.0.0.3	1	1	2026-09-11 11:32:43	2026-09-11 11:32:54
10.0.0.4	1	1	2026-09-11 11:22:38	2026-09-11 11:22:48

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-11 11:12:43	10.0.0.6	DDoS	UDP Flood	74.1%	High	Time Ban (1m)
2026-09-11 11:13:43	10.0.0.6	DDoS	UDP Flood	74.1%	Medium	Released
2026-09-11 11:14:16	10.0.0.6	DDoS	UDP Flood	74.1%	High	Quarantined
2026-09-11 11:15:56	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 11:15:56	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 11:15:57	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 11:16:06	10.0.0.20	DDoS	UDP Flood	75.8%	High	Time Ban (1m)
2026-09-11 11:16:06	10.0.0.21	DDoS	UDP Flood	78.1%	High	Time Ban (1m)
2026-09-11 11:16:07	10.0.0.20	DDoS	UDP Flood	75.8%	High	Time Ban
2026-09-11 11:16:07	10.0.0.21	DDoS	UDP Flood	78.1%	High	Time Ban
2026-09-11 11:16:08	10.0.0.22	DDoS	UDP Flood	77.7%	High	Time Ban (1m)
2026-09-11 11:16:08	10.0.0.22	DDoS	UDP Flood	77.7%	High	Time Ban
2026-09-11 11:17:07	10.0.0.20	DDoS	UDP Flood	77.6%	High	Time Ban (2m)
2026-09-11 11:17:07	10.0.0.21	DDoS	UDP Flood	78.1%	High	Time Ban (2m)
2026-09-11 11:17:08	10.0.0.22	DDoS	UDP Flood	86.0%	High	Released
2026-09-11 11:17:19	10.0.0.22	DDoS	UDP Flood	86.0%	Critical	Time Ban (2m)
2026-09-11 11:22:48	10.0.0.2	DDoS	Uncertain	59.0%	Low	Released
2026-09-11 11:22:48	10.0.0.4	DDoS	Uncertain	57.3%	Low	Released
2026-09-11 11:23:08	10.0.0.21	DDoS	UDP Flood	86.1%	High	Released
2026-09-11 11:28:21	10.0.0.22	DDoS	UDP Flood	77.7%	Critical	Blackhole
2026-09-11 11:32:35	10.0.0.21	DDoS	UDP Flood	78.3%	Critical	Blackhole
2026-09-11 11:32:54	10.0.0.3	DDoS	Uncertain	59.8%	Low	Released
2026-09-11 11:37:48	10.0.0.20	DDoS	UDP Flood	75.0%	Critical	Blackhole

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.20	UDP Flood	0.6330	75.0%	Critical	Phase 3	2	11s	2026-09-11 11:37:48	Escalated to Blackhole
10.0.0.20	UDP Flood	0.6330	75.0%	Critical	Phase 2	1	10s	2026-09-11 11:37:47	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6300	77.7%	Critical	Phase 3	1	1s	2026-09-11 11:37:41	Escalated to Blackhole
10.0.0.21	UDP Flood	0.6223	77.9%	Critical	Phase 3	1	1s	2026-09-11 11:37:41	Escalated to Blackhole
10.0.0.20	UDP Flood	0.6402	87.0%	Critical	Phase 2	2	1m 11s	2026-09-11 11:33:34	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.6	UDP Flood	0.5724	91.0%	Critical	Phase 1	6	11s	2026-09-11 11:33:00	Attack Stopped
10.0.0.3	Uncertain	0.6158	59.8%	Low	Phase 1	1	11s	2026-09-11 11:32:54	Attack Stopped
10.0.0.21	UDP Flood	0.6464	78.3%	Critical	Phase 3	2	11s	2026-09-11 11:32:35	Escalated to Blackhole
10.0.0.21	UDP Flood	0.6449	78.3%	Critical	Phase 2	1	10s	2026-09-11 11:32:34	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	75.9%	Critical	Phase 2	1	10s	2026-09-11 11:32:33	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6267	77.9%	Critical	Phase 3	1	1s	2026-09-11 11:32:25	Escalated to Blackhole
10.0.0.22	UDP Flood	0.6339	77.7%	Critical	Phase 2	1	1m 11s	2026-09-11 11:28:21	Ban Expired
10.0.0.21	UDP Flood	0.6464	87.4%	Critical	Phase 2	2	1m 10s	2026-09-11 11:28:21	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6395	79.5%	Critical	Phase 2	2	1m 10s	2026-09-11 11:28:20	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6398	77.7%	Critical	Phase 2	1	11s	2026-09-11 11:27:21	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6454	78.4%	Critical	Phase 2	1	10s	2026-09-11 11:27:21	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	75.6%	Critical	Phase 2	1	10s	2026-09-11 11:27:20	Attack persisted - escalating to time ban
10.0.0.6	UDP Flood	0.6093	92.0%	Critical	Phase 1	5	10s	2026-09-11 11:26:27	Attack Stopped
10.0.0.6	UDP Flood	0.6145	92.0%	High	Phase 1	4	11s	2026-09-11 11:25:47	Attack Stopped
10.0.0.22	UDP Flood	0.6398	86.3%	Critical	Phase 2	3	11s	2026-09-11 11:23:19	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6317	86.1%	Critical	Phase 2	3	11s	2026-09-11 11:23:19	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.5727	86.3%	High	Phase 2	1	1m 12s	2026-09-11 11:23:08	Ban Expired
10.0.0.21	UDP Flood	0.5549	86.1%	High	Phase 2	1	1m 12s	2026-09-11 11:23:08	Ban Expired
10.0.0.20	UDP Flood	0.6289	77.6%	High	Phase 2	2	1m 10s	2026-09-11 11:23:06	Attack persisted - escalating to time ban
10.0.0.2	Uncertain	0.6009	59.0%	Low	Phase 1	1	10s	2026-09-11 11:22:48	Attack Stopped
10.0.0.4	Uncertain	0.6098	57.3%	Low	Phase 1	1	10s	2026-09-11 11:22:48	Attack Stopped

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.22	UDP Flood	0.6398	75.0%	High	Phase 2	1	11s	2026-09-11 11:22:07	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6301	76.0%	High	Phase 2	1	11s	2026-09-11 11:22:07	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6330	76.2%	High	Phase 2	1	10s	2026-09-11 11:22:06	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6273	86.0%	Critical	Phase 2	3	11s	2026-09-11 11:17:19	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6106	86.0%	High	Phase 2	1	1m 11s	2026-09-11 11:17:08	Ban Expired
10.0.0.20	UDP Flood	0.6373	77.6%	High	Phase 2	2	1m 11s	2026-09-11 11:17:07	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6220	78.1%	High	Phase 2	2	1m 11s	2026-09-11 11:17:07	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6376	77.7%	High	Phase 2	1	11s	2026-09-11 11:16:08	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6336	75.8%	High	Phase 2	1	10s	2026-09-11 11:16:06	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6288	78.1%	High	Phase 2	1	10s	2026-09-11 11:16:06	Attack persisted - escalating to time ban
10.0.0.6	UDP Flood	0.6099	92.0%	High	Phase 1	3	10s	2026-09-11 11:15:06	Attack Stopped
10.0.0.6	UDP Flood	0.6155	74.1%	High	Phase 1	2	10s	2026-09-11 11:14:26	Attack Stopped
10.0.0.6	UDP Flood	0.5576	74.1%	Medium	Phase 2	1	1m 0s	2026-09-11 11:13:43	Ban Expired

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE